SECURITY ASSESSMENT REPORT

vulnerable-demo

Automated code, configuration and architecture audit

DATE
2026-11-13

RUN
#1

STRICTNESS
standard

SCOPE
vulnerable-demo

TOOL
PN-Ultra-Audit 1.0.34

CONFIDENTIAL

1 Executive summary

CRITICAL

Overall risk level

89 confirmed findings: 7 critical, 29 high, 32 medium and 20 low. The quality gate FAILS.

7	29	32	20
CRITICAL	HIGH	MEDIUM	LOW

SECURITY — Critical

E

A B C D **E**

RELIABILITY — Poor

D

A B C **D** E

MAINTAINABI... — Poor

D

A B C **D** E

Technical debt 26.8%

SEVERITY DISTRIBUTION



Critical	7	8%
High	29	33%
Medium	32	36%
Low	20	22%
Info	1	1%

OWASP TOP 10



KEY RISKS

SEVERITY	TITLE	FILE
CRITICAL	**Shell command execution (PHP)** Remote code execution on the server.	app/Http/UserController.php:10
CRITICAL	**Shell command execution (Python)** Remote code execution on the server.	backend/app.py:25
CRITICAL	**Shell command execution (Node)** Remote code execution on the server.	backend/server.js:6
CRITICAL	**Wallet private key hard-coded** Account/cloud/database takeover depending on the secret's scope.	contracts/.env.example2:1
CRITICAL	**Unsafe deserialization** Remote code execution.	backend/app.py:43

SEVERITY	TITLE	FILE
CRITICAL	**`initialize()` can be called by anyone and more than once** Irreversible loss of user funds or protocol control - contracts are public, immutable (or hard to upgrade) and attacked within minutes of deployment.	contracts/Vault.sol:26
CRITICAL	**1 of 2 sampled `users.password` values do not look hashed** A DB leak exposes every user's real password (and reuse elsewhere).	app.db:1
HIGH	**1 web-first assertion(s) not awaited - they never fail the test** Teams stop trusting CI, reruns hide real failures, and un-awaited assertions pass even when the page is broken.	web/tests/checkout.spec.ts:6
HIGH	**Android: debuggable / cleartext** Data extraction via adb/backup, MITM, component hijack, XSS-to-native bridges.	mobile/android/app/src/main/AndroidManifest.xml:2
HIGH	**Pages without <title> (1 of 1 HTML pages)** Lower rankings and click-through, poor social previews, duplicate-content confusion, weaker accessibility.	web/index.html:1

2 Scope and methodology

Target	vulnerable-demo
Project type	Full-stack (Backend/API + Frontend + Web + Mobile + DB + Smart contracts) — focus: Crypto / Web3 (high)
Auditors run	advisories, aitrace, db, docs, hardcode, malware, qa, quality, security, seo, tests, web3
Size	129 LOC · 22 files
Languages	Python, Solidity, JavaScript, PHP, TypeScript, Swift, SQL, Kotlin
Detected stack	Android, iOS, react, react-dom, Docker, Terraform
Strictness	standard

Methodology: static analysis (SAST) with untrusted-input tracing (taint), configuration, secrets, dependencies and SBOM, quality and architecture, tests, SEO/PWA and, when requested, databases and servers in read-only mode. No active exploitation was performed.

This report covers only what is listed under “Coverage”. No findings does not prove absence of defects: automatic rules are heuristics (regex + nearby variable tracking), not full data-flow analysis. “To review” items depend on context or on facts outside the repository. Verify each fix with its test before deploying.

3 Findings

SEVERITY	ID	TITLE	FILE	EFFORT
CRITICAL	SEC-002	**Shell command execution (PHP)**	app/Http/UserController.php:10	30 min
CRITICAL	SEC-003	**Shell command execution (Python)**	backend/app.py:25	30 min
CRITICAL	SEC-005	**Shell command execution (Node)**	backend/server.js:6	30 min
CRITICAL	SEC-006	**Wallet private key hard-coded**	contracts/.env.example2:1	30 min
CRITICAL	SEC-004	**Unsafe deserialization**	backend/app.py:43	45 min
CRITICAL	SEC-007	**`initialize()` can be called by anyone and more than once**	contracts/Vault.sol:26	45 min
CRITICAL	SEC-001	**1 of 2 sampled `users.password` values do not look hashed**	app.db:1	2.0 h
HIGH	BUG-002	**1 web-first assertion(s) not awaited - they never fail the test**	web/tests/checkout.spec.ts:6	3 min
HIGH	SEC-032	**Android: debuggable / cleartext**	mobile/android/app/src/main/AndroidManifest.xml:2	5 min
HIGH	SEO-001	**Pages without <title> (1 of 1 HTML pages)**	web/index.html:1	5 min
HIGH	SEO-002	**robots.txt blocks the whole site (Disallow: /)**	web/public/robots.txt:1	5 min
HIGH	SEC-009	**Request data echoed without encoding (PHP)**	app/Http/UserController.php:6	10 min
HIGH	SEC-015	**User input rendered as HTML without escaping (Python)**	backend/app.py:20	10 min
HIGH	SEC-019	**User input written straight into HTML response (Node)**	backend/server.js:4	10 min
HIGH	SEC-021	**User input written straight into HTML response (Node)**	backend/server.js:6	10 min
HIGH	BUG-001	**iOS: code uses a protected API but Info.plist lacks NSLocationWhenInUseUsageDescription (crash + App Store rejection)**	mobile/ios/App/Info.plist:1	10 min
HIGH	SEC-027	**tx.origin used for authorization (phishing via intermediate contract)**	contracts/Vault.sol:22	15 min
HIGH	SEC-031	**Raw HTML sink (innerHTML/document.write)**	frontend/src/App.jsx:2	15 min
HIGH	SEC-008	**SQL built with string concatenation (PHP)**	app/Http/UserController.php:5	20 min

SEVERITY	ID	TITLE	FILE	EFFORT
HIGH	SEC-010	**Object loaded by client-supplied id (check ownership)**	app/Http/UserController.php:7	20 min
HIGH	SEC-011	**Hard-coded credential**	backend/app.py:5	20 min
HIGH	SEC-012	**Database URL with credentials**	backend/app.py:6	20 min
HIGH	SEC-013	**SQL built with string formatting (Python)**	backend/app.py:12	20 min
HIGH	SEC-014	**SQL built with string formatting (Python)**	backend/app.py:19	20 min
HIGH	SEC-017	**File access with user-controlled path**	backend/app.py:39	20 min
HIGH	SEC-020	**SQL built with template/concatenation (JS/TS)**	backend/server.js:5	20 min
HIGH	SEC-022	**JWT signed with hard-coded secret**	backend/server.js:7	20 min
HIGH	SEC-023	**Object loaded by client-supplied id (check ownership)**	backend/server.js:8	20 min
HIGH	SEC-026	**Solidity tx.origin authorization**	contracts/Vault.sol:22	20 min
HIGH	SEC-029	**Over-broad GRANT**	db/schema.sql:3	20 min
HIGH	SEC-030	**Database user password in SQL script**	db/schema.sql:4	20 min
HIGH	SEC-016	**Outbound request to user-controlled URL**	backend/app.py:31	30 min
HIGH	SEC-018	**Password hashed with MD5/SHA-1**	backend/app.py:51	1.0 h
HIGH	SEC-024	**`withdraw()` updates state AFTER an external call (reentrancy)**	contracts/Vault.sol:12	1.0 h
HIGH	SEC-025	**`sweep()` moves value/mints/burns/selfdestructs but has no visible access control**	contracts/Vault.sol:18	1.0 h
HIGH	SEC-028	**`pickWinner()` derives randomness from block data (miner/validator-influenced)**	contracts/Vault.sol:24	2.0 h
MEDIUM	BUG-010	**Focused test committed (`.only` / `fit`) - the rest of the suite is silently skipped**	web/tests/checkout.spec.ts:8	2 min
MEDIUM	A11-001	**Image without alt text**	web/index.html:10	3 min
MEDIUM	SEC-033	**SQLite file is world-accessible (0o644)**	app.db:1	5 min
MEDIUM	SEC-038	**Secret/password written to logs**	frontend/src/App.jsx:4	5 min

SEVERITY	ID	TITLE	FILE	EFFORT
MEDIUM	TST-007	**package.json has no working `test` script**	package.json:1	5 min
MEDIUM	SEO-004	**Pages without meta description (1 of 1 HTML pages)**	web/index.html:1	5 min
MEDIUM	SEO-005	**Pages without viewport meta (not mobile-friendly) (1 of 1 HTML pages)**	web/index.html:1	5 min
MEDIUM	TST-005	**iOS: 1 sleep/asyncAfter calls in tests (flaky)**	mobile/ios/AppUITests/LoginUITests.swift:3	6 min
MEDIUM	TST-009	**Fixed sleeps in tests: 2 occurrence(s) in 1 file(s)**	web/tests/checkout.spec.ts:4	8 min
MEDIUM	BUG-006	**Loose comparison in auth/token check**	app/Http/UserController.php:9	10 min
MEDIUM	SEC-034	**Redirect to user-controlled URL**	backend/app.py:35	10 min
MEDIUM	SEC-035	**Non-cryptographic RNG for secrets**	backend/server.js:9	10 min
MEDIUM	A11-002	**Click handler on a non-interactive element (keyboard/screen-reader inaccessible)**	web/index.html:11	10 min
MEDIUM	BUG-007	**async callback in forEach (not awaited)**	backend/server.js:10	15 min
MEDIUM	COD-003	**Table `users` has no PRIMARY KEY**	db/schema.sql:1	20 min
MEDIUM	TST-003	**Test file `mobile/ios/AppUITests/LoginUITests.swift` defines tests but contains no assertions**	mobile/ios/AppUITests/LoginUITests.swift:1	20 min
MEDIUM	TST-008	**Test file `tests/test_app.py` defines tests but contains no assertions**	tests/test_app.py:1	20 min
MEDIUM	COD-002	**Table `users` has no PRIMARY KEY**	app.db:1	25 min
MEDIUM	SEO-003	**No sitemap.xml (or generator) found**	.:1	30 min
MEDIUM	BUG-003	**Invalid foreign key definition**	app.db:1	30 min
MEDIUM	BUG-004	**Money column `users.balance` uses REAL**	app.db:1	30 min
MEDIUM	BUG-005	**Money column `orders.total` uses REAL**	app.db:1	30 min
MEDIUM	BUG-008	**Money column `users.price` uses FLOAT**	db/schema.sql:1	30 min
MEDIUM	SEC-037	**Password column `users.password` too short for a modern hash (32)**	db/schema.sql:1	30 min
MEDIUM	BUG-009	**Money column `orders.total` uses FLOAT**	db/schema.sql:2	30 min
MEDIUM	COD-001	**Project has no README**	.:1	45 min
MEDIUM	SEC-036	**Solidity randomness from block data**	contracts/Vault.sol:24	1.0 h

SEVERITY	ID	TITLE	FILE	EFFORT
MEDIUM	TST-004	**iOS: UI tests exist but production code sets no accessibilityIdentifier**	mobile/ios/AppUITests/LoginUITests.swift:1	1.5 h
MEDIUM	TST-006	**Front-end has no unit/component test runner (Vitest/Jest/Karma)**	package.json:1	1.5 h
MEDIUM	SEM-001	**Tracking tags found (Google Analytics 4, Meta Pixel) but no consent management detected**	web/index.html:1	2.0 h
MEDIUM	TST-001	**Android: no unit tests (src/test) for the app code**	.:1	4.0 h
MEDIUM	TST-002	**Android: no instrumented/UI tests (src/androidTest)**	.:1	4.0 h
LOW	TST-012	**iOS: 1 UI test file(s) do not set continueAfterFailure = false**	mobile/ios/AppUITests/LoginUITests.swift:1	2 min
LOW	A11-003	**<html> without lang attribute**	web/index.html:2	2 min
LOW	COD-008	**Floating pragma (^0.8.0): deployed bytecode may differ from what was audited**	contracts/Vault.sol:1	5 min
LOW	DEP-001	**`package.json` has no lockfile**	package.json:1	5 min
LOW	SEO-006	**Pages without canonical link (1 of 1 HTML pages)**	web/index.html:1	5 min
LOW	SEO-007	**Pages with zero or multiple <h1> (1 of 1 HTML pages)**	web/index.html:1	5 min
LOW	SEO-008	**Pages missing Open Graph title/image (1 of 1 HTML pages)**	web/index.html:1	5 min
LOW	SEO-009	**Pages with ≥3 images lacking width/height (layout shift) (1 of 1 HTML pages)**	web/index.html:1	5 min
LOW	SEO-010	**Pages with render-blocking <script> (no async/defer) (1 of 1 HTML pages)**	web/index.html:1	5 min
LOW	DEP-002	**`web/package.json` has no lockfile**	web/package.json:1	5 min
LOW	TST-010	**iOS: code coverage not enabled in scheme/test plan**	.:1	10 min
LOW	COD-005	**Foreign keys declared but PRAGMA foreign_keys is OFF by default**	app.db:1	10 min
LOW	PER-001	**Foreign key `orders.user_id` without index**	app.db:1	10 min
LOW	BUG-011	**Mutable default argument that is mutated or leaked**	backend/app.py:53	10 min
LOW	COD-007	**Empty catch/except (swallowed error)**	backend/app.py:62	10 min
LOW	PER-002	**Foreign key `orders.user_id` without index**	db/schema.sql:2	10 min
LOW	COD-004	**Public API documentation coverage is 0.0% (target 60%)**	.:1	15 min

SEVERITY	ID	TITLE	FILE	EFFORT
LOW	COD-009	**Table `orders` uses MyISAM**	db/schema.sql:2	20 min
LOW	COD-006	**Public API under-documented in `backend/app.py` (0/12)**	backend/app.py:1	36 min
LOW	TST-011	**No Slither/Echidna/Medusa/Certora in CI**	.:1	1.5 h
INFO	SEO-011	**robots.txt does not reference the sitemap**	web/public/robots.txt:1	2 min

CRITICAL AND HIGH FINDINGS IN DETAIL

CRITICAL **SEC-002** CMD-PHP CWE-78 A03:2021

Shell command execution (PHP)

app/Http/UserController.php:10

Impact: Remote code execution on the server.

```
    $r = DB::select("SELECT * FROM users WHERE id = " . $id);
    echo "Hello " . $_GET['name'];
    $u = User::findOrFail($request->id);
    User::create($request->all());
    if ($password == $stored) { return true; }
    system("convert " . $_GET['f']);
  }
}

```

Remediation:

1. Avoid shelling out: use a library or native API for the job.
2. If unavoidable, pass an argument array with no shell (`subprocess.run([...], shell=False)`, `execFile`, `ProcessBuilder(list)`).
3. Validate arguments against an allowlist/regex; never build the command string from input.
4. Run the process with the least privileges / in a sandbox.

CRITICAL **SEC-003** CMD-PY CWE-78 A03:2021

Shell command execution (Python)

backend/app.py:25

Impact: Remote code execution on the server.

```
20      return render_template_string("<h1>Results for " + q + "</h1>")
21
22  @app.route("/ping")
23  def ping():
24      host = request.args.get("host")
25      os.system("ping -c 1 " + host)
26      return "ok"
27
28  @app.route("/fetch")
```

Remediation:

1. Avoid shelling out: use a library or native API for the job.
2. If unavoidable, pass an argument array with no shell (`subprocess.run([...], shell=False)`, `execFile`, `ProcessBuilder(list)`).
3. Validate arguments against an allowlist/regex; never build the command string from input.
4. Run the process with the least privileges / in a sandbox.

CRITICAL **SEC-005** CMD-JS CWE-78 A03:2021

Shell command execution (Node)

backend/server.js:6

Impact: Remote code execution on the server.

```
1  const express = require('express'); const jwt = require('jsonwebtoken'); const { exec } = require('child_process');
2  const app = express();
3  app.use(require('cors')({ origin: '*' }));
4  app.get('/hello', (req, res) => { res.send('<h1>Hello ' + req.query.name + '</h1>'); });
5  app.get('/u', async (req, res) => { const { id } = req.query; const r = await db.query(`SELECT * FROM users WHERE id = ${id}`); res.json(r);
   });
6  app.get('/run', (req, res) => { exec('ls ' + req.query.dir, (e, o) => res.send(o)); });
7  app.post('/login', (req, res) => { const t = jwt.sign({ u: req.body.u }, 'hardcodedjwtsecret'); res.json({ t }); });
8  app.get('/item/:id', async (req, res) => { const it = await Item.findById(req.params.id); res.json(it); });
9  const token = Math.random().toString(36); // reset token
```

Remediation:

1. Avoid shelling out: use a library or native API for the job.
2. If unavoidable, pass an argument array with no shell (`subprocess.run([...], shell=False)`, `execFile`, `ProcessBuilder(list)`).
3. Validate arguments against an allowlist/regex; never build the command string from input.
4. Run the process with the least privileges / in a sandbox.

CRITICAL **SEC-006** SEC-WALLET-PRIVKEY CWE-321 CWE-798 A02:2021

Wallet private key hard-coded

contracts/.env.example2:1

Impact: Account/cloud/database takeover depending on the secret's scope.

```
PRIV…
```

Remediation:

1. Wallet: treat it as stolen - create a new wallet and move all funds/approvals NOW (revoke token approvals on the old address); never reuse the seed phrase.
2. RPC/provider keys (Infura/Alchemy/QuickNode): regenerate in the provider dashboard and add domain/IP allow-lists.
3. ROTATE/REVOKE the secret now - assume it is compromised.
4. Move it to env vars / secret manager (Vault, AWS SM, GCP SM, Doppler, GitHub Secrets); read at runtime.

CRITICAL **SEC-004** DESER CWE-502 A08:2021

Unsafe deserialization

backend/app.py:43

Impact: Remote code execution.

```
38  def file():
39      return send_file("/var/data/" + request.args.get("f"))
40
41  @app.route("/load", methods=["POST"])
42  def load():
43      return str(pickle.loads(request.data))
44
45  @app.route("/page")
46  def page():
```

Remediation:

1. Use data-only formats (JSON) and validate with a schema.
2. `yaml.safe_load`, never `yaml.load` without SafeLoader.
3. If native serialization is unavoidable: sign with HMAC and verify before deserializing, and allowlist classes.
4. Never deserialize cookies/params/uploads from clients.

CRITICAL **SEC-007** WEB3-UNPROTECTED-INIT CWE-665

`initialize()` can be called by anyone and more than once

contracts/Vault.sol:26

Impact: Irreversible loss of user funds or protocol control - contracts are public, immutable (or hard to upgrade) and attacked within minutes of deployment.

```
21
22    function isOwner() public view returns (bool) { return tx.origin == owner; }
23
24    function pickWinner() external view returns (uint) { return uint(keccak256(abi.encodePacked(block.timestamp, msg.sender))) % 10; }
25
26    function initialize(address o) public { owner = o; }
27 }
28
```

Remediation:

1. Apply Checks-Effects-Interactions and OpenZeppelin `ReentrancyGuard`; update state before any external call.
2. Protect every state-changing/privileged function with explicit access control (Ownable2Step / AccessControl) and a multisig + timelock for admin actions.
3. Initializers: `initializer` modifier and `_disableInitializers()` in implementation constructors; never leave `initialize` callable.
4. Check every low-level call result; use SafeERC20 for tokens; avoid `tx.origin`; pin the compiler version (>= 0.8.x).

CRITICAL **SEC-001** DB-PLAINTEXT-PASSWORDS CWE-256 A02:2021

1 of 2 sampled `users.password` values do not look hashed

app.db:1

Impact: A DB leak exposes every user's real password (and reuse elsewhere).

```
1/2 sampled rows in users.password (values not shown)
```

Remediation:

1. Switch to argon2id (or bcrypt cost ≥ 12 / scrypt) via the framework (`password_hash`, `bcrypt`, `passlib`, Spring `BCryptPasswordEncoder`).
2. On next successful login, re-hash the password with the new algorithm (transparent migration); keep a `hash_version` column.
3. Force reset for dormant accounts after a grace period.

⚠ HIGH **BUG-002** QA-UNAWAITED-EXPECT CWE-754

1 web-first assertion(s) not awaited - they never fail the test

web/tests/checkout.spec.ts:6

Impact: Teams stop trusting CI, reruns hide real failures, and un-awaited assertions pass even when the page is broken.

```
1  import { test, expect } from '@playwright/test';
2  test('test1', async ({ page }) => {
3    await page.goto('/checkout');
4    await page.waitForTimeout(3000);
5    await page.locator('.btn-primary').click({ force: true });
6    expect(page.getByText('Done')).toBeVisible();
7  });
8  test.only('works', async ({ page }) => {
9    await page.goto('https://shop.example.com/cart');
```

Remediation:

1. Replace sleeps with auto-waiting web-first assertions (`await expect(locator).toBeVisible()`), `waitFor`, or explicit network/route waits.
2. Locate like a user: getByRole > getByLabel > getByText > getByTestId > CSS (last resort); avoid XPath and deep CSS chains.
3. Remove `force: true`: fix the real overlap/animation; always `await expect(...)`.
4. Make every test independent (own data, fresh context); avoid serial mode unless truly required.

⚠ HIGH **SEC-032** MOB-ANDROID-DEBUG CWE-489 CWE-319 M8:2024

Android: debuggable / cleartext

mobile/android/app/src/main/AndroidManifest.xml:2

Impact: Data extraction via adb/backup, MITM, component hijack, XSS-to-native bridges.

```
1  <manifest xmlns:android="http://schemas.android.com/apk/res/android">
2    <application android:debuggable="true" android:allowBackup="true" android:usesCleartextTraffic="true">
3      <activity android:name=".Main" android:exported="true"/>
4    </application>
5  </manifest>
6
```

Remediation:

1. Release builds: `android:debuggable=false`, `allowBackup=false` (or backup rules), `usesCleartextTraffic=false`.
2. Set `android:exported="false"` unless needed; protect exported components with permissions.
3. WebView: disable JS unless required, no `addJavascriptInterface` on untrusted content, no file access.
4. Store tokens in Keystore-backed `EncryptedSharedPreferences`/Keychain/`flutter_secure_storage`, not plain prefs/AsyncStorage.

⚠ HIGH **SEO-001** SEO-NO-TITLE

Pages without <title> (1 of 1 HTML pages)

web/index.html:1

Impact: Lower rankings and click-through, poor social previews, duplicate-content confusion, weaker accessibility.

```
1 <!doctype html>
2 <html>
3 <head>
4 <title></title>
5 <script src="/js/app.js"></script>
6 <script async src="https://www.googletagmanager.com/gtag/js?id=G-ABC123XYZ"></script>
```

Remediation:

1. Give every indexable page a unique <title> (30-60 chars, primary keyword first, brand last) and a unique meta description (70-160 chars, a reason to click).
2. Add <link rel="canonical"> pointing to the preferred absolute URL (self-referencing by default).
3. Exactly one <h1> per page that matches the search intent; use h2/h3 for structure.
4. Add Open Graph (og:title, og:description, og:image 1200x630, og:url) and twitter:card.

⚠ HIGH **SEO-002** SEO-ROBOTS-BLOCKS-ALL

robots.txt blocks the whole site (Disallow: /)

web/public/robots.txt:1

Impact: Pages are not indexed or waste crawl budget; link equity is lost; users hit errors.

```
1 User-agent: *
2 Disallow: /
3
```

Remediation:

1. Publish /robots.txt (allow CSS/JS, point to the sitemap) and /sitemap.xml containing only 200, indexable, canonical URLs; submit it in Search Console.
2. Fix 4xx/5xx pages and internal links to them; replace redirect chains with a single 301 to the final URL.
3. Force HTTPS and a single host (www vs apex) with 301s; add HSTS.
4. Never ship `noindex` on pages you want ranked (check staging flags leaking to production).

⚠ HIGH **SEC-009** XSS-ECHO-PHP CWE-79 A03:2021

Request data echoed without encoding (PHP)

app/Http/UserController.php:6

Impact: Session/token theft, actions as the victim, defacement, keylogging, pivot to admin accounts (stored XSS hits every viewer).

```
1  <?php
2  class UserController {
3    public function show() {
4      $id = $_GET['id'];
5      $r = DB::select("SELECT * FROM users WHERE id = " . $id);
6      echo "Hello " . $_GET['name'];
7      $u = User::findOrFail($request->id);
8      User::create($request->all());
9      if ($password == $stored) { return true; }
```

Remediation:

1. Identify the sink (innerHTML, v-html, dangerouslySetInnerHTML, `{!! !!}`, `|safe`, echo of request data).
2. Prefer auto-escaping output: `textContent`, `{{ }}`/`{{ }}` templates, `htmlspecialchars($x, ENT_QUOTES, 'UTF-8')`.
3. If HTML must be rendered (rich text), sanitize with an allowlist library (DOMPurify, sanitize-html, bleach, HTMLPurifier) at render time.
4. Validate URL schemes (`http`, `https`, `mailto` only) before putting user data in `href`/`src`.

⚠ HIGH **SEC-015** XSS-REFLECT-PY CWE-79 A03:2021

User input rendered as HTML without escaping (Python)

backend/app.py:20

Impact: Session/token theft, actions as the victim, defacement, keylogging, pivot to admin accounts (stored XSS hits every viewer).

```
15 @app.route("/search")
16 def search():
17     q = request.args["q"]
18     cur = sqlite3.connect("db/app.db").cursor()
19     cur.execute("SELECT * FROM items WHERE name = '%s'" % q)
20     return render_template_string("<h1>Results for " + q + "</h1>")
21
22 @app.route("/ping")
23 def ping():
```

Remediation:

1. Identify the sink (innerHTML, v-html, dangerouslySetInnerHTML, `{!! !!}`, `|safe`, echo of request data).
2. Prefer auto-escaping output: `textContent`, `{{ }}`/`{{ }}` templates, `htmlspecialchars($x, ENT_QUOTES, 'UTF-8')`.
3. If HTML must be rendered (rich text), sanitize with an allowlist library (DOMPurify, sanitize-html, bleach, HTMLPurifier) at render time.
4. Validate URL schemes (`http`, `https`, `mailto` only) before putting user data in `href`/`src`.

⚠ HIGH **SEC-019** XSS-REFLECT-JS CWE-79 A03:2021

User input written straight into HTML response (Node)

backend/server.js:4

Impact: Session/token theft, actions as the victim, defacement, keylogging, pivot to admin accounts (stored XSS hits every viewer).

```
1 const express = require('express'); const jwt = require('jsonwebtoken'); const { exec } = require('child_process');
2 const app = express();
3 app.use(require('cors')({ origin: '*' }));
4 app.get('/hello', (req, res) => { res.send('<h1>Hello ' + req.query.name + '</h1>'); });
5 app.get('/u', async (req, res) => { const { id } = req.query; const r = await db.query(`SELECT * FROM users WHERE id = ${id}`); res.json(r); });
6 app.get('/run', (req, res) => { exec('ls ' + req.query.dir, (e, o) => res.send(o)); });
7 app.post('/login', (req, res) => { const t = jwt.sign({ u: req.body.u }, 'hardcodedjwtsecret'); res.json({ t }); });
8 app.get('/item/:id', async (req, res) => { const it = await Item.findById(req.params.id); res.json(it); });
9 const token = Math.random().toString(36); // reset token
```

Remediation:

1. Identify the sink (innerHTML, v-html, dangerouslySetInnerHTML, `{!! !!}`, `|safe`, echo of request data).
2. Prefer auto-escaping output: `textContent`, `{{ }}`/`{{ }}` templates, `htmlspecialchars($x, ENT_QUOTES, 'UTF-8')`.
3. If HTML must be rendered (rich text), sanitize with an allowlist library (DOMPurify, sanitize-html, bleach, HTMLPurifier) at render time.
4. Validate URL schemes (`http`, `https`, `mailto` only) before putting user data in `href`/`src`.

⚠ HIGH **SEC-021** XSS-REFLECT-JS CWE-79 A03:2021

User input written straight into HTML response (Node)

backend/server.js:6

Impact: Session/token theft, actions as the victim, defacement, keylogging, pivot to admin accounts (stored XSS hits every viewer).

```
1 const express = require('express'); const jwt = require('jsonwebtoken'); const { exec } = require('child_process');
2 const app = express();
3 app.use(require('cors')({ origin: '*' }));
4 app.get('/hello', (req, res) => { res.send('<h1>Hello ' + req.query.name + '</h1>'); });
5 app.get('/u', async (req, res) => { const { id } = req.query; const r = await db.query(`SELECT * FROM users WHERE id = ${id}`); res.json(r); });
6 app.get('/run', (req, res) => { exec('ls ' + req.query.dir, (e, o) => res.send(o)); });
7 app.post('/login', (req, res) => { const t = jwt.sign({ u: req.body.u }, 'hardcodedjwtsecret'); res.json({ t }); });
8 app.get('/item/:id', async (req, res) => { const it = await Item.findById(req.params.id); res.json(it); });
9 const token = Math.random().toString(36); // reset token
```

Remediation:

1. Identify the sink (innerHTML, v-html, dangerouslySetInnerHTML, `{!! !!}`, `|safe`, echo of request data).
2. Prefer auto-escaping output: `textContent`, `{{ }}`/`{{ }}` templates, `htmlspecialchars($x, ENT_QUOTES, 'UTF-8')`.
3. If HTML must be rendered (rich text), sanitize with an allowlist library (DOMPurify, sanitize-html, bleach, HTMLPurifier) at render time.
4. Validate URL schemes (`http`, `https`, `mailto` only) before putting user data in `href`/`src`.

⚠ HIGH **BUG-001** MOB-IOS-PERM-MISSING CWE-276 M6:2024

iOS: code uses a protected API but Info.plist lacks NSLocationWhenInUseUsageDescription (crash + App Store rejection)

mobile/ios/App/Info.plist:1

Impact: Android: SecurityException/feature silently fails. iOS: the app crashes on first use and fails App Store review.

```
1  <?xml version="1.0"?><plist version="1.0"><dict><key>CFBundleName</key><string>App</string></dict></plist>
2
```

Remediation:

1. Android: add `<uses-permission android:name="..."/>` and request it at runtime before use.
2. iOS: add the matching `NS...UsageDescription` key to Info.plist with a clear, user-facing reason.
3. Test the permission-denied path too (no crash, helpful UI).

⚠ HIGH **SEC-027** WEB3-TX-ORIGIN CWE-477

tx.origin used for authorization (phishing via intermediate contract)

contracts/Vault.sol:22

Impact: Irreversible loss of user funds or protocol control - contracts are public, immutable (or hard to upgrade) and attacked within minutes of deployment.

```
17
18     function sweep(address payable to) external {
19         to.transfer(address(this).balance);
20     }
21
22     function isOwner() public view returns (bool) { return tx.origin == owner; }
23
24     function pickWinner() external view returns (uint) { return uint(keccak256(abi.encodePacked(block.timestamp, msg.sender))) % 10; }
25
```

Remediation:

1. Apply Checks-Effects-Interactions and OpenZeppelin `ReentrancyGuard`; update state before any external call.
2. Protect every state-changing/privileged function with explicit access control (Ownable2Step / AccessControl) and a multisig + timelock for admin actions.
3. Initializers: `initializer` modifier and `_disableInitializers()` in implementation constructors; never leave `initialize` callable.
4. Check every low-level call result; use SafeERC20 for tokens; avoid `tx.origin`; pin the compiler version (>= 0.8.x).

⚠ HIGH **SEC-031** XSS-DOM CWE-79 A03:2021

Raw HTML sink (innerHTML/document.write)

frontend/src/App.jsx:2

Impact: Session/token theft, actions as the victim, defacement, keylogging, pivot to admin accounts (stored XSS hits every viewer).

```
export default function App({ html, user }) {
  useEffect(() => { document.getElementById('x').innerHTML = location.hash; }, []);
  localStorage.setItem('token', user.token);
  console.log('login', user.password);
  return <div dangerouslySetInnerHTML={{ __html: html }} />;
}

```

Remediation:

1. Identify the sink (innerHTML, v-html, dangerouslySetInnerHTML, `{!! !!}`, `|safe`, echo of request data).
2. Prefer auto-escaping output: `textContent`, `{{ }}`/`{{ }}` templates, `htmlspecialchars($x, ENT_QUOTES, 'UTF-8')`.
3. If HTML must be rendered (rich text), sanitize with an allowlist library (DOMPurify, sanitize-html, bleach, HTMLPurifier) at render time.
4. Validate URL schemes (`http`, `https`, `mailto` only) before putting user data in `href`/`src`.

⚠ HIGH **SEC-008** SQLI-PHP CWE-89 A03:2021

SQL built with string concatenation (PHP)

app/Http/UserController.php:5

Impact: Read, modify or delete any table (including password hashes), bypass login, and on some engines execute OS commands.

```
<?php
class UserController {
  public function show() {
    $id = $_GET['id'];
    $r = DB::select("SELECT * FROM users WHERE id = " . $id);
    echo "Hello " . $_GET['name'];
    $u = User::findOrFail($request->id);
    User::create($request->all());
    if ($password == $stored) { return true; }
```

Remediation:

1. Find every place the query string is built (the flagged line and its callers).
2. Replace concatenation/interpolation with placeholders (`?`, `%s`, `$1`, `:name`) and pass values separately.
3. For values that cannot be parameters (ORDER BY column, table name, sort direction) map user input to a fixed allowlist.
4. Give the application DB user least privilege (no DROP/ALTER/FILE, only needed tables).

⚠ HIGH **SEC-010** AUTHZ-IDOR CWE-639 CWE-862 A01:2021 API1:2023

Object loaded by client-supplied id (check ownership)

app/Http/UserController.php:7

Impact: Cross-user and cross-tenant data exposure - the most common high-impact web bug.

```
2  class UserController {
3    public function show() {
4      $id = $_GET['id'];
5      $r = DB::select("SELECT * FROM users WHERE id = " . $id);
6      echo "Hello " . $_GET['name'];
7      $u = User::findOrFail($request->id);
8      User::create($request->all());
9      if ($password == $stored) { return true; }
10     system("convert " . $_GET['f']);
```

Remediation:

1. Scope the lookup through the owner: `$user->orders()->findOrFail($id)` / `Order.objects.get(pk=id, owner=request.user)`.
2. Or authorize explicitly (`Gate::authorize('view', $order)`, policies, `@PreAuthorize`).
3. Use a global tenant scope; add a test where user B requests user A's id and expects 403/404.

⚠ HIGH **SEC-011** SEC-HARDCODED CWE-798 A07:2021

Hard-coded credential

backend/app.py:5

Impact: Account/cloud/database takeover depending on the secret's scope.

```
SECRET_KEY = "supe…"
```

Remediation:

1. Generic secret: identify which service issues it, rotate it there, and load the new value from the environment or a secret manager.
2. Search the codebase and history for the same value (`git grep`, `gitleaks detect`) - it may be duplicated.
3. ROTATE/REVOKE the secret now - assume it is compromised.
4. Move it to env vars / secret manager (Vault, AWS SM, GCP SM, Doppler, GitHub Secrets); read at runtime.

⚠ HIGH **SEC-012** SEC-DB-URL CWE-798 A07:2021

Database URL with credentials

backend/app.py:6

Impact: Account/cloud/database takeover depending on the secret's scope.

```
DB_URL = "post…"
```

Remediation:

1. Database: change the password/user NOW (`ALTER USER ... PASSWORD ...`), update the app's secret store, and restrict network access (private network / IP allow-list).
2. Review connections and recent queries/audit logs; if data may have been read, follow your breach process.
3. ROTATE/REVOKE the secret now - assume it is compromised.
4. Move it to env vars / secret manager (Vault, AWS SM, GCP SM, Doppler, GitHub Secrets); read at runtime.

⚠ HIGH **SEC-013** SQLI-PY CWE-89 A03:2021

SQL built with string formatting (Python)

backend/app.py:12

Impact: Read, modify or delete any table (including password hashes), bypass login, and on some engines execute OS commands.

```
7
8  @app.route("/users")
9  def users():
10     sort = request.args.get("sort")
11     cur = sqlite3.connect("db/app.db").cursor()
12     cur.execute(f"SELECT * FROM users ORDER BY {sort}")
13     return str(cur.fetchall())
14
15 @app.route("/search")
```

Remediation:

1. Find every place the query string is built (the flagged line and its callers).
2. Replace concatenation/interpolation with placeholders (`?`, `%s`, `$1`, `:name`) and pass values separately.
3. For values that cannot be parameters (ORDER BY column, table name, sort direction) map user input to a fixed allowlist.
4. Give the application DB user least privilege (no DROP/ALTER/FILE, only needed tables).

⚠ HIGH **SEC-014** SQLI-PY CWE-89 A03:2021

SQL built with string formatting (Python)

backend/app.py:19

Impact: Read, modify or delete any table (including password hashes), bypass login, and on some engines execute OS commands.

```
14
15  @app.route("/search")
16  def search():
17      q = request.args["q"]
18      cur = sqlite3.connect("db/app.db").cursor()
19      cur.execute("SELECT * FROM items WHERE name = '%s'" % q)
20      return render_template_string("<h1>Results for " + q + "</h1>")
21
22  @app.route("/ping")
```

Remediation:

1. Find every place the query string is built (the flagged line and its callers).
2. Replace concatenation/interpolation with placeholders (`?`, `%s`, `$1`, `:name`) and pass values separately.
3. For values that cannot be parameters (ORDER BY column, table name, sort direction) map user input to a fixed allowlist.
4. Give the application DB user least privilege (no DROP/ALTER/FILE, only needed tables).

⚠ HIGH **SEC-017** PATH-TRAVERSAL CWE-22 A01:2021

File access with user-controlled path

backend/app.py:39

Impact: Read secrets/source (`/etc/passwd`, `.env`) or overwrite files.

```
34  def go():
35      return redirect(request.args.get("next"))
36
37  @app.route("/file")
38  def file():
39      return send_file("/var/data/" + request.args.get("f"))
40
41  @app.route("/load", methods=["POST"])
42  def load():
```

Remediation:

1. Never use the raw name. Map an ID to a stored path, or generate random server-side file names.
2. Resolve the final path and verify it stays under the base directory (`realpath(base + name).startswith(realpath(base))`).
3. Reject `..`, absolute paths, null bytes; use framework helpers (`send_from_directory`, `Storage::disk`).
4. For archives, check each entry's resolved path before extracting (zip-slip).

4 Remediation roadmap

NOW (CRITICAL/HIGH)	THIS SPRINT	BACKLOG
36	**50**	**3**
~17.0 h	~23.6 h	~2.1 h

5 Repository structure

22 files · 99 findings

```
├── backend/  [27 · critical]
│   ├── app.py  [18 · critical]
│   └── server.js  [9 · critical]
├── contracts/  [10 · critical]
│   ├── Vault.sol  [9 · critical]
│   └── .env.example2  [1 · critical]
├── app/  [6 · critical]
│   └── Http/  [6 · critical]
│       └── UserController.php  [6 · critical]
├── web/  [19 · high]
│   ├── index.html  [13 · high]
│   ├── tests/  [3 · high]
│   │   └── checkout.spec.ts  [3 · high]
│   ├── public/  [2 · high]
│   │   └── robots.txt  [2 · high]
│   └── package.json  [1 · low]
├── db/  [11 · high]
│   └── schema.sql  [11 · high]
├── mobile/  [11 · high]
│   ├── android/  [6 · high]
│   │   └── app/  [6 · high]
│   └── ios/  [5 · high]
│       ├── App/  [1 · high]
│       └── AppUITests/  [4 · medium]
├── frontend/  [4 · high]
│   └── src/  [4 · high]
│       └── App.jsx  [4 · high]
├── package.json  [5 · medium]
├── tests/  [1 · medium]
│   └── test_app.py  [1 · medium]
├── Dockerfile  [3]
├── infra/  [1]
│   └── main.tf  [1]
└── .env  [1]
```

6 Appendix: coverage

Coverage

covered **12** deferred **4**

UNIT	PASS	STATUS	NOTES
source scan (regex + taint look-back)	audit-security,audit-bugs	Covered	22 files; truncated rules: none
documentation / standards / architecture	audit-code	Covered	strictness=standard

UNIT	PASS	STATUS	NOTES
test-suite analysis (static)	audit-quality	Covered	file-name matching; coverage % not measured unless --run-tests
QA front-end + mobile (static)	audit-qa,audit-mobile-qa	Covered	mobile=android, ios
static SEO / SEM	audit-seo	Covered	templates, robots, sitemap, metadata, tracking/consent
repository hygiene (keys, history, CI/CD)	audit-security	Covered	sensitive files in git/history, remote credentials, workflow security, SECURITY.md/CODEOWNERS/dependabot/secret scanning
malware & backdoor indicators	audit-security	Covered	webshells, reverse shells, decode-and-execute, miners, stealers, install hooks, persistence, polyglots, Trojan Source (static, offline; not an antivirus)
version advisories (OSV · CISA KEV · NVD · endoflife.date)	audit-security	Covered	dependency CVEs by locked version, runtime/framework end-of-life, CVEs of the exact runtime, recent actively exploited bugs of the stack
hardcode & TODOs	audit-quality	Covered	TODO/FIXME/HACK (age via git blame), security TODOs, remove-before-prod markers, stubs, environment URLs, IPs, machine paths, personal emails, identity checks, business rates
AI traces (project finish)	audit-docs	Covered	AI comments/docs, agent & spec files, git trailers, provenance metadata, invisible characters

NOTES

- db-fixes.sql (suggested indexes/constraints to REVIEW; never executed by the tool) and db-schema.mmd (Mermaid ER diagram) were generated.
- repo-fixes.sh was generated (guided repository clean-up; the tool never runs it): `bash repo-fixes.sh list`.
- fix-plan.md was generated: 117 findings in 102 groups and 5 phases (phase 0 comes first).

This report covers only what is listed under “Coverage”. No findings does not prove absence of defects: automatic rules are heuristics (regex + nearby variable tracking), not full data-flow analysis. “To review” items depend on context or on facts outside the repository. Verify each fix with its test before deploying.